Practice workbook · Two's complement numbers

8-bit two's complement integers: pattern to denary, negative denary to pattern, and negating a pattern.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T3-03

1. Question 3 · part b.i · 2 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part b.i only.

3 Computers use binary to represent and store data.

(a) Two images are stored.

- Image 1 has a colour depth of 8 bits.
- Image 2 has a colour depth of 16 bits.

(i) State the number of colours that can be represented in Image 1.

(1)

.....

(ii) Image 2 is 1920 pixels wide by 1080 pixels high.

Construct an expression to show the file size of the image in mebibytes.

You do **not** need to do the calculation.

(3)

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(iii) One pixel in Image 1 is represented by the unsigned binary integer

1100 1010

Identify the denary equivalent.

(1)

- ☒ **A** 54
- ☒ **B** 191
- ☒ **C** 202
- ☒ **D** –74

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1. Question 3 · part b.i · 2 marks

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Worked-example retry · Attempt part b.i only.

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(b) A register holds the value 1100 0001 as an 8-bit binary value in two's complement.

(i) Convert the two's complement binary integer 1100 0001 to denary.

(2)

(ii) Give the result of applying an arithmetic shift left by one to the binary pattern 1100 0001

(1)

(iii) The value 1111 1111 is added to the value 1100 0001 stored in the register.

An error occurs.

Explain this error.

(2)

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1. Question 3 · part b.i · 2 marks · Working and answer

Working and answer

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2. Question 2 · part a.ii · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part a.ii only.

2 Computers use binary digits to represent data.

(a) Sign and magnitude and two's complement are two ways of representing signed integer numbers.

(i) Convert the denary number -94 to a binary pattern using sign and magnitude representation.

(2)

(ii) 1000 1001 is a signed integer that uses two's complement representation.

Convert this signed integer to denary.

(2)

(b) Denary numbers use base 10

Identify the base for hexadecimal numbers.

(1)

- ☒ **A** 32
- ☒ **B** 16
- ☒ **C** 8
- ☒ **D** 2

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2. Question 2 · part a.ii · 2 marks · Working and answer

Working and answer

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3. Question 1 · part a.ii · 2 marks

2021 NOV · Paper 01

Further practice · Attempt part a.ii only.

1 Computers use binary to represent and store data.

(a) Binary and hexadecimal notation can be used to represent numbers.

(i) Convert the denary number 77 to 8-bit binary.

(2)

(ii) Convert the denary number –126 to 8-bit binary using two's complement.

(2)

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(iii) Convert the binary pattern 11000110 to hexadecimal.

(2)

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3. Question 1 · part a.ii · 2 marks · Working and answer

Working and answer

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4. Question 2 · part d.iii · 2 marks

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Further practice · Attempt part d.iii only.

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2 Computers manipulate binary patterns. People interpret those patterns.

(a) Complete the table by adding the hexadecimal notation for each of the denary values.

(2)

Denary	Hexadecimal
8	
12	

(b) Identify the expression to give the number of unique binary patterns that can be stored in six bits.

(1)

- ☒ **A** 6^2
- ☒ **B** 6×2
- ☒ **C** $6^2 - 1$
- ☒ **D** 2^6

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(c) Binary patterns are manipulated by shifts.

(i) Give the result of applying a logical shift right by two to the binary pattern 0101 1100

(1)

(ii) Give the result of applying an arithmetic shift right by three to the binary pattern 1100 0101

(1)

4. Question 2 · part d.iii · 2 marks

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Further practice · Attempt part d.iii only.

(d) Binary patterns can be interpreted as signed or unsigned integers.

(i) Convert the denary unsigned integer 60 to 8-bit binary.

(2)

(ii) Here is a binary bit pattern for a signed integer in sign and magnitude format.

1001 0110

Convert the binary bit pattern to denary.

Be sure to include a sign symbol in your answer.

(2)

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6

P 7 2 9 3 7 A 0 6 2 4

4. Question 2 · part d.iii · 2 marks

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Further practice · Attempt part d.iii only.

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(iii) Negating a signed integer means changing its sign without changing its value.

The negation of +16 is –16. The negation of –24 is +24.

Here is the binary bit pattern for a signed integer in two's complement format.

1110 0101

Convert the binary pattern to its negation in two's complement.

(2)

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(e) Construct an expression to convert 13 kilobytes to kibibytes.

You **do not** need to do the calculation.

(2)

(Total for Question 2 = 13 marks)

4. Question 2 · part d.iii · 2 marks · Working and answer

Working and answer

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